SK-2N01G Shale Density Meter

Technical Datasheet | Manual hydrometer-type instrument for mud logging formation-pressure evaluation

Purpose

The SK-2N01G shale density meter determines the apparent bulk density of selected shale or mudstone cuttings at the wellsite. Repeated density measurements plotted against depth can support recognition of departures from a normal compaction trend and may contribute to abnormal formation-pressure evaluation.

The instrument is a manual comparative weighing system based on Archimedes' principle. It produces two scale readings for the same specimen: one in air and one submerged in fresh water.



Key specifications

Item	Specification
Model	SK-2N01G (also referenced as 2N01G in legacy catalog records)
Measurement range	1.00-3.00 g/cm³
Minimum sample mass	0.05 g
Resolution / sensitivity	0.03 g/cm³
Overall size	Approx. Ø130 × 380 mm; confirm against supplied unit
Measurement method	Two-reading hydrostatic weighing in clean fresh water
Calculation	Density = L1 / (L1 - L2)
Power requirement	None; manual instrument
Primary application	Wellsite shale/mudstone cuttings density trending

Instrument assembly

- Transparent measuring cylinder and protective cover.
- Mirror-backed graduated scale and fixed scale support.
- Stainless-steel rod with upper sample pan.
- Immersed float / lower specimen support.
- Stabilizing weight and zero-adjustment control.

Operating principle

The first reading, L1, represents the response with the prepared specimen on the upper pan in air. The second reading, L2, is obtained after moving the same specimen to the submerged support. The difference (L1 - L2) represents the buoyancy-related response associated with displaced water volume. The ratio of the air response to this difference gives the specimen density.

$$\text{Density} = L1 / (L1 - L2)$$

Performance depends strongly on representative sample selection, consistent washing and blotting, removal of visible bubbles, stable water temperature, correct zero adjustment and reading the scale only after the assembly has stabilized.

Recommended sample and use conditions

- Use representative, competent shale or mudstone cuttings from a known depth interval.
- Avoid obvious cavings, highly porous or gumbo shale, cracked pieces, contaminating minerals and flat fragments that trap air or behave inconsistently.
- Wash gently with the fluid defined by the site program, then blot surface liquid with absorbent paper. Measure promptly and use the same preparation routine for every interval.
- Keep the cylinder vertical on a stable, vibration-free bench and prevent the stabilizing weight or moving assembly from touching the cylinder bottom or wall.

Data use and limitations

A single value is not a pressure diagnosis. Interpret shale density as a depth trend together with lithology, sample quality, lag correction, drilling parameters, gas, resistivity, sonic or other available formation-pressure indicators. A decrease or reduced rate of density increase may indicate undercompaction, but mineralogy, hydration, contamination, cavings and sample handling can produce similar effects.

This instrument is not a certified safety device and must not be used as the sole basis for well-control decisions. Follow the operator's approved well-control and formation-evaluation procedures.

Ordering and verification

Before purchase or field use, confirm the model, scale range, supplied cylinder, float assembly, upper and lower specimen supports, stabilizing weight, zero-adjustment mechanism, protective cover and calibration / inspection documentation. Dimensions and accessories may vary by production batch.

Source basis

Prepared from the ChinaMudLogging.com legacy product record, public SK-2N01G product specifications, and published mud logging operating guidance. Product values should be confirmed against the supplier's current quotation and the delivered instrument.

References: https://www.ca800.com/news/d_1nrusj6oanl78.html | https://imudlogging.com/product/iml-3dt01-shale-density-meter | https://www.chinamudlogging.com/product/shale-density-meter-for-mud-logging-unit/

Document revision: 1.0 | Date: 2026-10-14